

Advanced Modeling

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Necessary Packages

```
library(tidyverse)
library(rsample)
library(caret)
library(vip)
library(ROCR)
library(nnet)
```

Necessary Data Sets

```
setwd('/Users/camsmithers/Desktop/Camalytics/NBA')
teamvisdata <- readRDS('Data-NBA/Current/teamvisdata.rds')
teamvisdata_fixed <- na.omit(teamvisdata) %>%
  mutate(
    team = as.factor(team),
    opponent = as.factor(opponent),
    season = as.factor(season))
```

Linear Regression Model Testing: Player Statistics

Logistic Regression Model Testing: Binary Game Outcome

```
winbinary_split <- initial_split(teamvisdata_fixed, prop=.75, strat='twoleveloc')
winbinary_train <- training(winbinary_split)
winbinary_test <- testing(winbinary_split)
```

For the models I started by adding all the variables that I wanted and deciding which ones I wanted to remove to see how the accuracy changes.

```
winbinary_1 <- train(
  twoleveloc ~ roll_offrating + roll_defrating + team + homeaway +
  vorp_bpa + roll_three_par + roll_ftar + roll_ast_pct,
  data=teamvisdata_fixed,
  method='glm',
  family='binomial',
  trControl=trainControl(method='cv', number=10))
winbinary_2 <- train(
  twoleveloc ~ roll_offrating + roll_defrating,
  data=teamvisdata_fixed,
```

```

method='glm',
family='binomial',
trControl=trainControl(method='cv', number=10))
winbinary_3 <- train(
  twoleveloc ~ roll_offrating + opponent,
  data=teamvisdata_fixed,
  method='glm',
  family='binomial',
  trControl=trainControl(method='cv', number=10))
winbinary_4 <- train(
  twoleveloc ~ homeaway,
  data=teamvisdata_fixed,
  method='glm',
  family='binomial',
  trControl=trainControl(method='cv', number=10))

```

This will give me summary statistics of the models' accuracy.

```

summary(
  resamples(
    list(
      model1=winbinary_1,
      model2=winbinary_2,
      model3=winbinary_3,
      model4=winbinary_4
    )
  )
)$statistics$Accuracy

```

	##	Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NA's
model1	##	0.6491345	0.6561876	0.6596806	0.6638257	0.6713240	0.6866267	0
model2	##	0.6380572	0.6533026	0.6602335	0.6582380	0.6626747	0.6739854	0
model3	##	0.6021291	0.6248545	0.6316129	0.6281606	0.6368929	0.6440452	0
model4	##	0.5395875	0.5487359	0.5558882	0.5545660	0.5581434	0.5692410	0

ROC Curve for Logistic Regression Models

Creating a plot to show the False Positive vs. True Positive Accuracy.

```

model1_prob <- predict(winbinary_1, winbinary_train, type='prob')$Win
perf1 <- prediction(model1_prob, winbinary_train$twoleveloc) %>%
  performance(measure='tpr', x.measure='fpr')

model2_prob <- predict(winbinary_2, winbinary_train, type='prob')$Win
perf2 <- prediction(model2_prob, winbinary_train$twoleveloc) %>%
  performance(measure='tpr', x.measure='fpr')

model3_prob <- predict(winbinary_3, winbinary_train, type='prob')$Win
perf3 <- prediction(model3_prob, winbinary_train$twoleveloc) %>%
  performance(measure='tpr', x.measure='fpr')

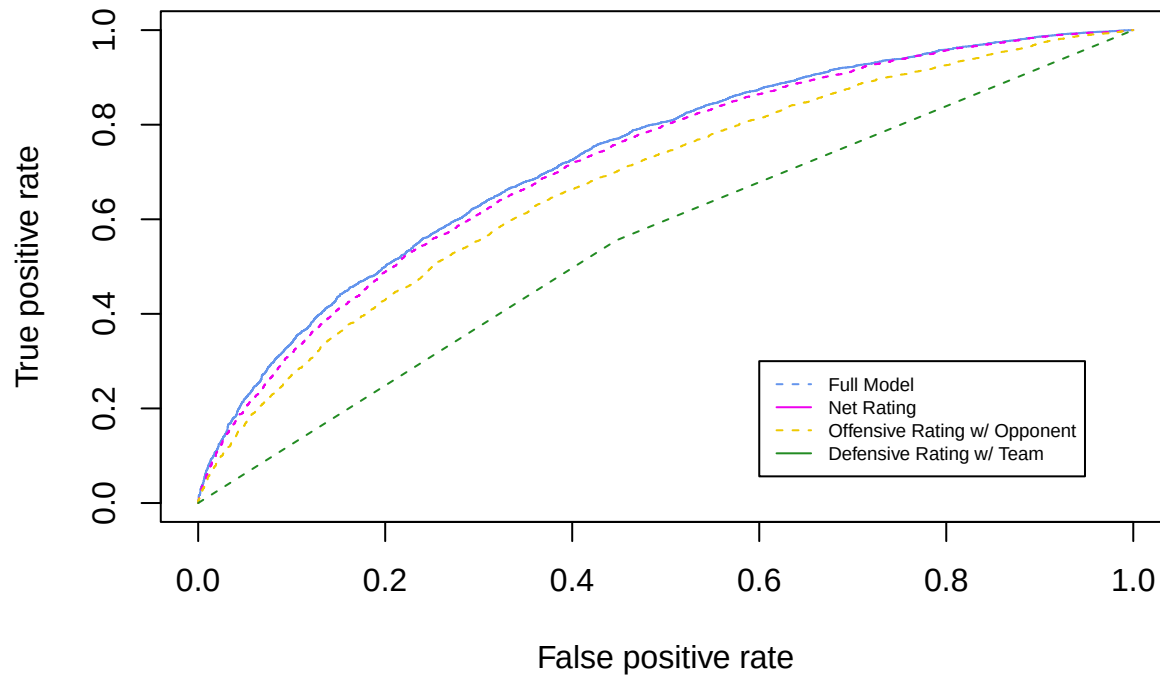
model4_prob <- predict(winbinary_4, winbinary_train, type='prob')$Win
perf4 <- prediction(model4_prob, winbinary_train$twoleveloc) %>%
  performance(measure='tpr', x.measure='fpr')

```

```

plot(perf1, col='cornflowerblue', lty=1)
plot(perf2, add=TRUE, col='magenta2', lty=2)
plot(perf3, add=TRUE, col='gold2', lty=2)
plot(perf4, add=TRUE, col='forestgreen', lty=2)
legend(0.6, 0.3, legend=c('Full Model',
                          'Net Rating',
                          'Offensive Rating w/ Opponent',
                          'Defensive Rating w/ Team'),
      col=c('cornflowerblue', 'magenta2', 'gold2', 'forestgreen'),
      lty=2:1, cex=0.6)

```



Multinomial Regression Testing: Outcome w/ Magnitude